

Relative Exponential Periods on Polynomial Chains: Algebraic Independence over Fixed Boundary Fields

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Abstract

We construct rational differential systems for compact polynomial-chain exponential integrals, with polynomial twisted de Rham representatives and literal Stokes forcing from fixed faces. For real algebraic phases of coprime weighted type, irreducibility and actual-chain detection give linear comparison modulo the boundary span, and scalar transcendence at every nonzero algebraic parameter. No critical-difference separation or boundary field disjointness is needed for this first conclusion. Detection is supplied by a saddle, a continued density germ, or a strict interior extremum on a positive domain, including an extremum of value zero.

With separated ordered critical-value differences and adjoint boundary disjointness, the $(a - 1)(b - 1)$ monomial moments are algebraically independent over the fixed-face field, both functionally and at every nonzero algebraic parameter. Finite root separation replaces rational independence and holds on a nonempty Zariski-open subset of the full weighted coefficient family. Spectral and rank tests certify boundary disjointness. Every polynomial amplitude has a unique Laurent Stokes remainder; its specialized vanishing characterizes membership in the boundary span at the linear level and algebraicity over the boundary field at the stronger level.

The proof uses Picard–Lefschetz geometry, Fourier irreducibility, a separated-root differential-Galois argument, and stable lattices. We retain the elliptic and rank-six triangles and add a directly proved noncoprime rank-three example with phase $x^4 + y^2 + x$. All boundary fields use only the specified faces. Unrestricted simplex nonvanishing, several-chain comparison, and general formal completeness are not asserted.

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1 Introduction and two levels of comparison

We study actual compact integrals and their polynomial Stokes certificates, not an unspecified completion of a formal period algebra. The general boundary construction leads to two distinct conclusions: linear comparison modulo the actual boundary span, and algebraic independence over its fraction field under additional hypotheses. Keeping these conclusions separate makes both the weaker assumptions and the stronger conclusion explicit. The elliptic, rank-six, and noncoprime rank-three triangles are worked applications.

The algebraic changes of variables, linearity, and twisted Stokes identities used here belong to the exponential Kontsevich–Zagier framework [6, 5]. The conclusions below concern one actual column over its fixed boundary field. They do not assert a presentation of all formal exponential-period relations. The functional and numerical independence proofs do not use the companion formal-comparison theorems or Paper IV.

Structure and previous version. Section 2 constructs actual systems and their E -function coordinates. Section 3 gives the linear criterion; Section 4 upgrades to boundary-field independence. Later sections prove irreducibility, root separation, boundary certificates, and actual detection, then specialize at every nonzero algebraic parameter. Section 8 proves the precise Zariski-open

phase statement; the real-geometric chain conditions are not part of that assertion. Section 14 retains the explicit continuation needed for the original triangle. The pre-integration version [9], including its separate observable-Lefschetz, Hodge, face-lattice, and multicolumn discussions, is archived unchanged. Those additional discussions are not inputs to the proof here. The earlier unrestricted endpoint-complete claim remains withdrawn; it is not restored by this theorem.

Revision. The October 2026 two-level revision adds linear comparison without root or boundary separation, the positive-domain extremum detector, finite root separation and its Zariski-open locus, and the noncoprime rank-three application. It retains the 6 October single-chain proof and both of its examples. The author list and original month-year date are unchanged. The factorial-moment applications are developed separately in Paper VI [8]; the present proofs do not depend on that paper.

Throughout $k = \mathbb{Q}$, $k_{\mathbb{R}} = k \cap \mathbb{R}$, $F_0 = k(z)$, $F = \mathbb{C}(z)$.

Definition 1.1 (Polynomial two-chain). A polynomial two-chain over $k_{\mathbb{R}}$ is a finite sum

$$D = \sum_{\ell=1}^N a_{\ell}(\Phi_{\ell})_*[0, 1]^2, \quad a_{\ell} \in k, \quad \Phi_{\ell} \in k_{\mathbb{R}}[u, v]^2,$$

with the standard orientation on each square. The restrictions of the Φ_{ℓ} to the oriented edges of the squares are the *faces* of D ; $|\partial D|$ denotes the union of their images in $\mathbb{R}^2$. Cancellations between coincident oriented faces are allowed.

For a polynomial phase q and a polynomial two-chain D , put

$$I_D(\omega)(z) = \int_D e^{zq} \omega, \quad d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

$K_{\partial,0}$ is the differential field over F_0 generated by the polynomial-amplitude integrals over the fixed faces of D and the exponentials at their endpoints; $K_{\partial} = \mathbb{C}K_{\partial,0}$; $K_{\partial,\xi}$ is the field generated over k by the values at ξ . For a face e with parametrization $\phi_e : [a_e, b_e] \rightarrow \mathbb{R}^2$ put $p_e = q \circ \phi_e$ and $d_e = \deg p_e$.

Definition 1.2 (The class (W)). Fix coprime integers $a, b \geq 2$ and give x, y the weights $\text{wt}(x) = b$, $\text{wt}(y) = a$. A phase is of class (W) if

$$q = \alpha x^a + \beta y^b + q_{<}, \quad \alpha\beta \neq 0, \quad q_{<} \in k_{\mathbb{R}}[x, y] \text{ of weighted degree } < ab,$$

with $\alpha, \beta \in k_{\mathbb{R}}$. Put $\mu = (a-1)(b-1)$ and $B = \{x^i y^j : 0 \leq i \leq a-2, 0 \leq j \leq b-2\}$.

The elliptic phase $y^2 - x^3 + 3x$ is of class (W) with $(a, b) = (3, 2)$, $\mu = 2$.

Definition 1.3 (Chain multiplicity). Let $D = \sum_{\ell} a_{\ell}(\Phi_{\ell})_*[0, 1]^2$ be a polynomial two-chain. For $P \notin |\partial D|$ define

$$m_D(P) = \sum_{\ell} a_{\ell} \deg(\Phi_{\ell}, (0, 1)^2, P),$$

where the last factor is the Brouwer degree. Thus m_D is a compactly supported, locally constant k -valued function on $\mathbb{R}^2 \setminus |\partial D|$. We call $m_D(P)$ the *multiplicity of D at P* . This definition includes non-injective parametrizations.

Lemma 1.4 (Phase densities). *Let $q \in k_{\mathbb{R}}[x, y]$ have finite critical locus, let D be a polynomial two-chain over $k_{\mathbb{R}}$, and let $r \in k[x, y]$.*

- (a) *There is a compactly supported L^1 function $\rho_{D,r} : \mathbb{R} \rightarrow \mathbb{C}$, unique almost everywhere, such that*

$$I_D(r \, dx \wedge dy)(z) = \int_{\mathbb{R}} e^{zt} \rho_{D,r}(t) \, dt.$$

At every regular value t it has the coarea representative

$$\rho_{D,r}(t) = \int_{q^{-1}(t)} \frac{m_D(x, y) r(x, y)}{\|\nabla q(x, y)\|} d\mathcal{H}^1.$$

On an interval containing no critical value of q , no critical value of a face phase p_e on its parameter interval, and no vertex value, this representative is real analytic; its germ therefore has a holomorphic complexification.

- (b) *For a face moment*

$$J_{e,h}(z) = \int_{a_e}^{b_e} h(s) e^{z p_e(s)} \, ds, \quad h \in k[s],$$

and an interval avoiding the critical values and endpoint values of p_e , the inverse-Laplace density is algebraic over $\mathbb{C}(t)$. More precisely, after splitting into monotone inverse branches $s_j(t)$ it is

$$\rho_{e,h}(t) = \sum_j \frac{h(s_j(t))}{|p'_e(s_j(t))|} = \sum_j \varepsilon_j \frac{h(s_j(t))}{p'_e(s_j(t))}, \quad \varepsilon_j \in \{\pm 1\}.$$

Endpoint exponentials (and the constant function 1) are Dirac point masses at their phase values.

Proof. For each summand of D , the degree formula for a map of the square gives

$$\int_D r e^{zq} \, dx \, dy = \int_{\mathbb{R}^2} m_D(x, y) r(x, y) e^{zq(x, y)} \, dx \, dy.$$

This also proves directly that non-injective parametrizations are counted with their signed local degrees. Since m_D has compact support, the coarea formula applied to q gives (a), including L^1 -integrability. Away from the stated exceptional values, the implicit-function theorem makes the regular level arcs and their intersections with the polynomial faces vary real analytically. Splitting the compact support into the finitely many resulting analytic pieces and integrating in analytic coordinates gives the asserted real-analytic representative.

For (b), split $[a_e, b_e]$ into monotonicity intervals for p_e . Ordinary one-variable change of variables gives the displayed formula. Each inverse branch $s_j(t)$ is algebraic, and the sign of p'_e is constant on the branch, so the density is algebraic. A value supported at an endpoint is the stated Dirac mass. $\square$

Whenever we speak below of the density germ of $I_D(r \, dx \wedge dy)$, we mean the analytic germ of the representative in Lemma 1.4(a).

Hypotheses. Let q be of class (W) and let D be a polynomial two-chain over $k_{\mathbb{R}}$ in the sense of Definition 1.1. This arithmetic restriction is used in (iii).

- (M) *Morse.* q has μ distinct critical values $c_1, \dots, c_\mu$ (equivalently: all critical points are nondegenerate with distinct values).

- (R) *Separated root weights.* All ordered differences $c_i - c_j$, $i \neq j$, are distinct: equality $c_i - c_j = c_k - c_l$ with $i \neq j$ and $k \neq l$ implies $(i, j) = (k, l)$.
- (F) *Faces.* For every face e with $d_e \geq 2$, the set $\{c_i - c_j : i \neq j\}$ is not contained in $\{u - v : u, v \text{ critical values of } p_e\}$.
- (S) *Saddle.* Some real critical point P_0 of q with $\det \text{Hess } q(P_0) < 0$ lies off the image of ∂D , the multiplicity $w := m_D(P_0)$ is nonzero, and $c := q(P_0)$ is neither the value of q at a vertex of D nor a critical value of any p_e on its parameter interval.
- (E) *Extremum.* D represents the usual positive area measure on the closure of a bounded real domain with nonempty interior, and

$$\min_D q < \min_{|\partial D|} q \quad \text{or} \quad \max_D q > \max_{|\partial D|} q.$$

Here $|\partial D|$ is the union of all specified face images, including any internal faces in the chosen presentation. This condition is not asserted for arbitrary signed or complex chains.

The older condition (Q), rational independence of $c_2 - c_1, \dots, c_\mu - c_1$, implies (R); it is no longer assumed. For $\mu = 2$, (R) is automatic.

Let $\mathcal{B} = (B_j)_j$ be the finite closed vector of reduced fixed-face moments, endpoint exponentials, and 1 supplied by Theorem 2.1. Set

$$\mathcal{N} = \sum_j F B_j, \quad \mathcal{V}_{\partial, \xi} = \sum_j k B_j(\xi).$$

The latter is a vector space, not the field $K_{\partial, \xi}$.

Theorem 1.5 (Linear single-chain comparison). *Let q satisfy (W), (M), and let the polynomial two-chain D over $k_{\mathbb{R}}$ satisfy (S), (S*) below, or (E). Write $M_m = I_D(m dx \wedge dy)$ for $m \in B$. Integration induces an injective differential map $H_q^2 \otimes_{F_0} F \longrightarrow \mathcal{M}/\mathcal{N}$. For every $\xi \in k^\times$,*

$$\sum_{m \in B} \lambda_m M_m(\xi) \in \mathcal{V}_{\partial, \xi}, \quad \lambda_m \in k \quad \implies \quad \lambda_m = 0 \text{ for all } m.$$

For $r \in k[x, y]$, let $a_{r, m}$ be the unique Laurent Stokes remainder coefficients of Lemma 5.1. Then

$$\begin{aligned} I_D(r dx \wedge dy)(\xi) \in \mathcal{V}_{\partial, \xi} &\iff a_{r, m}(\xi) = 0 \quad \text{for every } m \in B \\ &\iff r dx \wedge dy \in (d + \xi dq \wedge) \Omega_{k[x, y]/k}^1. \end{aligned}$$

In particular every nonzero $r \in \text{span}_k B$ has a transcendental integral at every nonzero algebraic parameter. Neither (R), (Q), nor (F) is required.

Remark 1.6. For a general polynomial amplitude, the specialized remainder may vanish. That puts its integral in the boundary span; the boundary term need not be zero. A nonzero remainder gives transcendence over k , not necessarily over the boundary field. The following theorem has that stronger conclusion.

Theorem 1.7 (Single-chain comparison). *Let q be of class (W) satisfying (M), (R), and let D be a polynomial two-chain over $k_{\mathbb{R}}$ satisfying one of (S), (S*), (E). Assume (F), or the intrinsic adjoint exclusion described in Corollary 9.5; its rank test also suffices. Put $M_m = I_D(m dx \wedge dy)$ for $m \in B$.*

- (i) The classes of $m dx \wedge dy$, $m \in B$, form an F_0 -basis of H_q^2 , and a $k[z, z^{-1}]$ -basis of the lattice of polynomial forms. Every $r \in k[x, y]$ has a unique remainder and a polynomial Stokes certificate

$$r = \sum_{m \in B} a_{r,m}(z) m + L_x P_r + L_y Q_r, \quad a_{r,m} \in k[z^{-1}], \quad P_r, Q_r \in z^{-1} k[z^{-1}][x, y],$$

with $L_x = \partial_x + zq_x$, $L_y = \partial_y + zq_y$.

- (ii) The μ functions M_m are algebraically independent over K_∂ .
- (iii) For every $\xi \in k^\times$ the μ numbers $M_m(\xi)$ are algebraically independent over $K_{\partial, \xi}$.
- (iv) For $r \in k[x, y]$ and $\xi \in k^\times$, the value $I_D(r dx \wedge dy)(\xi)$ is algebraic over $K_{\partial, \xi}$ iff it lies in $K_{\partial, \xi}$ iff $r dx \wedge dy \in (d + \xi dq \wedge) \Omega_{k[x, y]/k}^1$.

Remark 1.8 (The continuation alternative). The saddle detector in Lemma 10.1 may be replaced by:

- (S*) on some interval $e_0 \subset \mathbb{R}$ free of critical values of q and of the p_e , of vertex values and of 0, the density germ $\rho = \rho_{D,1}$ of $I_D(dx \wedge dy)$ from Lemma 1.4 continues analytically along a path to a punctured neighbourhood of some c_i , where its variation around c_i is a nonzero constant multiple of the vanishing period $\int_{\nu_i(t)} dx \wedge dy/dq$.

Lemma 14.1 verifies (S*) for the original elliptic triangle: its saddle $(1, 0)$ lies outside the triangle, so (S) itself fails. Section 14 verifies (F) and derives its functional and numerical comparison from Theorem 1.7.

Corollary 1.9 (Polygons). *Let $\min(a, b) \geq 3$, let q be of class (W) with (M), (R), and let D be a real polygonal chain (all faces straight segments) with vertices in $k_{\mathbb{R}}^2$. Then (F) holds automatically. If moreover the coefficients of q and the vertices lie in a field $k_0 \subset k_{\mathbb{R}}$ over which $[k_0(c_i) : k_0] \geq \max(a, b)$ for all i , then (S) holds as soon as some real saddle of q lies in the interior of D with nonzero multiplicity.*

Proof. A straight face has $d_e \leq \max(a, b)$, so p_e has at most $\max(a, b) - 1 < \mu$ critical values and at most $(\mu - 1)(\mu - 2) < \mu(\mu - 1)$ nonzero differences, whereas under (R) the $\mu(\mu - 1)$ numbers $c_i - c_j$ are pairwise distinct. Vertex values lie in k_0 and critical values of $p_e \in k_0[s]$ have degree at most $d_e - 1 < \max(a, b)$ over k_0 , so neither can equal c . $\square$

Example 1.10 (A rank-six instance). For

$$q = x^4 + y^3 + xy - y, \quad D = \text{conv}\{(0, -1), (1, -1), (\tfrac{1}{2}, 0)\},$$

the six functions $\int_D x^i y^j e^{zq} dx dy$ ($0 \leq i \leq 2$, $0 \leq j \leq 1$) are algebraically independent over the field of the three edges, and so are their values at every nonzero algebraic ξ over the edge-value field. See §13.

Differential-module convention. For a system $Y' = EY$ we use its coefficient module: a column of basis elements e satisfies $\partial e = Ee$. Solutions are differential homomorphisms out of this module, so the solution functor is contravariant. Its horizontal representation is dual to the solution representation. The natural trace duality identifies the corresponding endomorphism and trace-free endomorphism representations. All module embeddings below are under the covariant horizontal-fibre equivalence with these identifications.

2 Polynomial representatives and literal boundary forcing

Put $k = \overline{\mathbb{Q}}$ and $F_0 = k(z)$. Let D be a polynomial two-chain over $k_{\mathbb{R}}$ as in Definition 1.1, and let $q \in k[x, y]$. For a polynomial two-form ω , write

$$I_D(\omega) = \int_D e^{zq} \omega.$$

When coefficients of ω lie in F_0 , interpret this as a meromorphic function of z by linearity. For each oriented face parametrization $\phi_e : [a_e, b_e] \rightarrow \mathbb{A}^2$, put $p_e = q \circ \phi_e$. The actual boundary field $K_{\partial,0}$ is the differential field over F_0 generated by all functions

$$\int_{a_e}^{b_e} h(t) e^{z p_e(t)} dt, \quad h \in k[t],$$

and the exponential functions at their algebraic endpoints. Only the fixed faces are used; this does not mean the field of all compact line periods.

Let d differentiate only in x, y , and set

$$d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

Theorem 2.1 (Polynomial-representative boundary system). *The space H_q^2 is finite dimensional. There are forms $\omega_i = r_i(x, y) dx \wedge dy$, with $r_i \in k[x, y]$, whose classes form an F_0 -basis, a matrix $A \in M_n(F_0)$, and polynomial-in- x, y one-forms $\beta_i \in \Omega_{F_0[x,y]/F_0}^1$ such that*

$$q\omega_i = \sum_j A_{ij}(z)\omega_j + d_z \beta_i. \quad (2.1)$$

The actual integral column $U_i = I_D(\omega_i)$ satisfies

$$U' = AU + b, \quad b_i = \int_{\partial D} e^{zq} \beta_i \in K_{\partial,0}. \quad (2.2)$$

Every polynomial two-form ω has an identity

$$\omega = \sum_i a_i(z)\omega_i + d_z \beta, \quad (2.3)$$

where $a_i \in F_0$ and β is polynomial in x, y . Consequently

$$I_D(\omega) = \sum_i a_i(z)U_i + \int_{\partial D} e^{zq} \beta. \quad (2.4)$$

For any finite set of amplitudes, these functions belong to a finite closed rational first-order system whose coordinates are actual E -functions. No spatial denominator is introduced. Parameter denominators may occur.

Proof. The polynomial Gauss–Manin complex for $q : \mathbb{A}^2 \rightarrow \mathbb{A}_t^1$ is

$$(\Omega_{k[x,y]/k}^{\bullet+2}[\partial_t], d - dq \wedge \partial_t),$$

and its cohomology is holonomic [10, Proposition 9.1]. The Fourier substitution $\partial_t \mapsto -z$, $t \mapsto \partial_z$ turns its differential into $d + z dq \wedge$. Fourier transform is exact on Weyl modules, and localization

to $k(z)$ is exact. The degree-zero cohomology of the localized transformed complex is therefore precisely H_q^2 , which is finite dimensional over $k(z)$ by holonomicity. This uses neither cohomological tameness nor the rank and lattice conclusions stated later in [10]. In particular, no identification with a Milnor number is asserted here.

The monomial two-forms with coefficients in k span the quotient over F_0 , so a finite subset is a basis. The connection on the quotient is

$$\nabla_z[\omega] = [\partial_z\omega + q\omega],$$

because

$$[\partial_z + q, d + z dq\wedge] = dq \wedge - dq\wedge = 0.$$

Expressing $q\omega_i$ in the basis proves (2.1); expressing an arbitrary class proves (2.3).

For a one-form polynomial in the spatial variables, there is the literal identity

$$e^{zq}d_z\beta = d_{x,y}(e^{zq}\beta).$$

Stokes on the original parametrized compact chain proves (2.2) and (2.4). Each pullback of β to a face is a finite F_0 -linear combination of polynomial one-forms in its parameter. Thus every forcing term is an actual face integral with coefficients in F_0 .

To see finiteness on the boundary, if $\deg p = d \geq 2$, division by $\partial_t + zp'(t)$ reduces every polynomial amplitude to degree at most $d - 2$. The leading degree of $(\partial_t + zp')h$, for $h \neq 0$, is $\deg h + d - 1$, with nonzero leading coefficient in F_0 . The remainder is therefore unique. Integration of the exact part gives only endpoint exponentials. Multiplication by p and the same division give a rational first-order system for the reduced moments and the endpoint exponentials. For $d = 1$ the twisted operator is surjective on polynomials; for $d = 0$ ordinary polynomial antiderivatives suffice. In both cases only endpoint terms remain. For finitely many faces, this supplies a finite boundary vector; its fraction field is $K_{\partial,0}$.

The integral coordinates are entire and have algebraic Taylor coefficients. After pullback to the square they have the form $\int_{[0,1]^2} P(u,v)e^{zQ(u,v)} du dv$. For the first m Taylor coefficients, one denominator of the form

$$d^{m+1} \text{lcm}(1, \dots, Cm + C_0)^2$$

suffices, with fixed integers d, C, C_0 ; all conjugates grow exponentially. The same argument in one variable handles the faces. Together with the constructed differential systems, these are the E -function conditions. $\square$

Proposition 2.2 (Actual affine comparison under a change of representatives). *Suppose $\tilde{\omega} = T(z)\omega + d_z\gamma$, with $T \in \text{GL}_n(F_0)$. Put $g = \int_{\partial D} e^{zq}\gamma$. Then the actual columns satisfy*

$$\tilde{U} = TU + g,$$

and their systems are related by

$$\tilde{A} = (T' + TA)T^{-1}, \quad \tilde{b} = Tb + g' - \tilde{A}g.$$

In particular, this is a comparison of the actual affine solution objects, not a statement only about their homogeneous composition factors.

Proof. The first equality is Stokes, and differentiating it proves the other two. If two primitives give the same form identity, their difference is d_z -closed and its integral over ∂D is zero by the same Stokes identity. Thus primitive choices do not introduce an arbitrary constant. $\square$

After adjoining complex constants, write $F = \mathbb{C}(z)$, $K_\partial = \mathbb{C}K_{\partial,0}$, and work in a field $\mathcal{M}$ of meromorphic germs containing the actual functions. Integration modulo the actual boundary field gives a canonical differential map

$$K_\partial \otimes_{F_0} H_q^2 \longrightarrow \mathcal{M}/K_\partial, \quad [\omega] \longmapsto I_D(\omega) \bmod K_\partial. \quad (2.5)$$

Here $\mathcal{M}/K_\partial$ is used as a differential vector space, not as a quotient field or ring. This map need not be injective for arbitrary chains. The stronger criterion, Theorem 4.3, proves injectivity rather than building it into a completion definition.

3 Linear comparison modulo the boundary span

Proposition 3.1 (Actual linear criterion). *Suppose an actual column U satisfies $U' = AU + b$ over F , where each b_i lies in a finite-dimensional differential subspace $\mathcal{N} \subset \mathcal{M}$. If the coefficient module of A is irreducible and $U \notin \mathcal{N}^n$, the map $e_i \mapsto U_i \bmod \mathcal{N}$ is injective. Thus the coordinates of U are F -linearly independent modulo $\mathcal{N}$. If, moreover, the combined boundary-and-interior column is a vector of E -functions in a closed system over $k[z, z^{-1}]$, its values at every $\xi \in k^\times$ are k -linearly independent modulo the boundary-value span.*

Proof. The displayed assignment is a differential map because the forcing lies in $\mathcal{N}$. Its kernel is a differential submodule. Irreducibility and the nonzero-image hypothesis force the kernel to vanish. There is no scalar extension from F to the fraction field $F(\mathcal{N})$ in this argument.

For the value statement, apply Beukers' linear lifting theorem [2, Theorem 3.2, p. 375] to the entire combined vector, including any redundant boundary coordinates. At the ordinary point $\xi \neq 0$ every algebraic linear relation is the specialization of a rational functional relation regular at ξ . Equivalently, Lemma 3.1 of that paper supplies polynomial relation rows with full rank on specialization. A functional relation has every interior coefficient identically zero by the first part, hence also zero at ξ . This proves the assertion for the prescribed relation, without clearing a denominator that vanishes at ξ . $\square$

Lemma 3.2 (Strict interior extremum). *Suppose q is real, D satisfies (E), and the twisted module is irreducible of rank at least two. Integration modulo $\mathcal{N}$ is injective. In particular $I_D(dx \wedge dy) \notin \mathcal{N}$ whenever $[dx \wedge dy] \neq 0$.*

Proof. Consider the minimum case, and put $c = \min_D q < d = \min_{|\partial D|} q$. First omit the artificially adjoined constant 1 and let $\mathcal{N}_b$ be the span of the genuine face moments and endpoint exponentials. It is a differential subspace: differentiation and one-variable Stokes reduction introduce only these same coordinates. For every $g \in \mathcal{N}_b$ there are C, N such that, for all sufficiently large real s ,

$$|g(-s)| \leq Cs^N e^{-ds}.$$

Rational coefficients have at most polynomial growth on that ray away from their finitely many poles. On the other hand, a positive-area sublevel neighborhood gives, for $0 < \varepsilon < d - c$,

$$I_D(dx \wedge dy)(-s) \geq A_\varepsilon e^{-(c+\varepsilon)s}, \quad A_\varepsilon > 0.$$

Thus the integration map modulo $\mathcal{N}_b$ is nonzero and hence injective. Its image is an irreducible module of rank at least two. Its intersection with $(\mathcal{N}_b + F)/\mathcal{N}_b$, a differential module of rank at most one, is zero. Passing to the quotient by this latter module restores 1 without losing injectivity. This includes $c = 0$, and also covers the case when an actual boundary coordinate already equals 1. The maximum case uses the positive ray, or equivalently replaces q by $-q$ and z by $-z$. $\square$

The hypothesis concerns the specified face support. Without positivity, several chain contributions near an extremum can cancel, so the preceding argument is not an extremum test for arbitrary signed chains.

4 An actual-solution comparison criterion

Let $\mathcal{B}$ be a finite rational boundary differential system, let L_∂/F be a Picard–Vessiot field for that system containing its actual solution coordinates, and let $\mathcal{N} \subset L_\partial$ be their finite-dimensional F -linear differential span. Include 1 when needed and put $K_\partial = F(\mathcal{N}) \subset L_\partial$. All fields are embedded in a common no-new-constants differential overfield. Let $\mathcal{T}_\partial$ be the tensor category generated by $\mathcal{B}$, allowing subquotients and duals. Its objects are precisely the differential modules trivialized by L_∂ [11, Theorem 1.5.2].

Lemma 4.1 (Differentially finite elements of a Picard–Vessiot field). *Every finite-dimensional F -differential subspace of L_∂ belongs to $\mathcal{T}_\partial$.*

Proof. Let R be the Picard–Vessiot ring. If $f \in L_\partial$ satisfies a scalar linear equation over F , all its Galois translates satisfy that equation. Their constant span is finite dimensional. After trivializing the torsor, a rational function with finite-dimensional translation orbit has no pole: the union of the polar loci of a finite basis is a proper closed subset invariant under all translations, whereas the group acts transitively on itself. This remains valid component by component if the group is disconnected. Thus $f \in R$; see the locally finite argument in [11, proof of Proposition 1.3.33].

Write $R = F[Y_{ij}, (\det Y)^{-1}]$ for a boundary fundamental matrix. A bounded span of monomials in Y_{ij} and $(\det Y)^{-1}$ is stable under differentiation, since $Y' = A_\partial Y$ and $((\det Y)^{-1})' = -\operatorname{tr}(A_\partial)(\det Y)^{-1}$. Such a span is a quotient of finite sums of tensor constructions in the boundary module and its determinant dual. Every finite-dimensional F -differential subspace of R lies in one of these spans. Closure under subquotients and the Picard–Vessiot tensor equivalence [11, Theorem 1.5.2] prove the claim. $\square$

Lemma 4.2 (Regular affine cocycles). *Let G be an algebraic group over an algebraically closed field C_0 of characteristic zero, with reductive identity component, and let V be a finite-dimensional rational representation. Every regular cocycle $c(gh) = c(g) + gc(h)$ has the form $c(g) = v - gv$ for some $v \in V$.*

Proof. Let g act on $V \oplus C_0$ by $g(w, t) = (gw + tc(g), t)$. Complete reducibility supplies an invariant lift $(v, 1)$ of 1 in the trivial quotient, giving the assertion. Complete reducibility here includes disconnected groups: average a G° -equivariant splitting over the finite component group [7, Corollary 22.43]. Equivalently, this is the vanishing of algebraic $H^1(G, V)$ in [7, Proposition 15.15]. $\square$

Theorem 4.3 (Actual one-column comparison). *Suppose an actual column U satisfies $U' = AU + b$, with $b_i \in \mathcal{N}$, and $n \geq 2$. Assume:*

1. *the homogeneous Galois group over F contains SL_n ;*
2. *its adjoint module $\operatorname{End}^0(A)$ does not belong to $\mathcal{T}_\partial$;*
3. *at least one coordinate of U is not in $\mathcal{N}$.*

Then $U_1, \dots, U_n$ are algebraically independent over $K_\partial \subset L_\partial$. In the setting of Theorem 2.1, (2.5) is injective, and its image is the span of the actual polynomial-amplitude periods modulo K_∂ .

Proof. Let $E = \mathcal{N} + \sum_i F U_i$. The differential module $E/\mathcal{N}$ is a quotient of the coefficient module associated with A . Use the coefficient module with basis e_i and $\partial e_i = \sum_j A_{ij} e_j$; the map $e_i \mapsto U_i \bmod \mathcal{N}$ is a differential map. Its horizontal representation is dual to the solution representation of $Y' = AY$, so it is irreducible and has the same adjoint tensor module. By the third assumption, the quotient is nonzero and hence is that entire irreducible module.

If all U_i belonged to L_∂ , Lemma 4.1 would put E , then $E/\mathcal{N}$, and then its adjoint module in $\mathcal{T}_\partial$, contradicting the second assumption. Thus $U \notin L_\partial^n$.

In a joint Picard–Vessiot extension for A and the boundary system, the kernel of restriction to L_∂ maps to a normal subgroup H of the homogeneous group $G \subset \mathrm{GL}_n$. A normal subgroup of a group containing SL_n either contains SL_n or has trivial projective image. Indeed, if $H \cap \mathrm{SL}_n$ is not all of SL_n , it is finite central. For each $h \in H$, the commutator map from the connected group SL_n to this finite intersection is constant and equals the identity at 1. Thus h centralizes SL_n and is scalar. This also excludes a hidden noncentral finite projective image. In the latter, projectively trivial case, H acts trivially on $\mathrm{End}^0(A)$, so the adjoint module is trivialized by L_∂ , a contradiction. Therefore the homogeneous group after base change to L_∂ , and hence to K_∂ , contains SL_n .

Let Y be a fundamental matrix over a joint field for the inhomogeneous system over K_∂ . Its affine Galois group is a subgroup of $\mathbb{C}^n \rtimes G_{K_\partial}$, acting by $\sigma(U) = U + Y c_\sigma$. Its translation kernel is an invariant vector subspace, hence either 0 or $\mathbb{C}^n$. If it is $\mathbb{C}^n$, translations make the orbit of U Zariski dense in $\mathbb{A}^n$. If it is zero, the remaining regular algebraic cocycle of the homogeneous group is a coboundary by Lemma 4.2. Here reductivity of its identity component follows also directly: a unipotent radical would lie in SL_n and be normal under SL_n , which has no nontrivial connected normal unipotent subgroup. Explicitly, writing $\sigma(Y) = Y g_\sigma$, the cocycle can be written $c_\sigma = g_\sigma c - c$; hence $U - Y c$ is fixed. Thus

$$U = a + Y c, \quad a \in K_\partial^n, \quad c \in \mathbb{C}^n.$$

The vector c is nonzero, since otherwise $U \in K_\partial^n$. The SL_n -orbit of a nonzero vector is $\mathbb{A}^n \setminus \{0\}$, again dense. In either case a polynomial over K_∂ vanishing at U vanishes on a dense affine orbit and is zero. Linear independence modulo the boundary field, and then the last assertion by normal-form reduction, follow immediately. $\square$

Remark 4.4. The third condition concerns only rational *linear* combinations of a finite actual boundary vector. It is strictly weaker than the desired nonmembership in the boundary fraction field. The proof is what upgrades this linear test to the fraction-field statement. A nonzero abstract cohomology class alone would not suffice.

5 Normal form for the class (W)

Lemma 5.1. *Let q be of class (W) (coprimality is not used here).*

- (a) *Every $r \in k[x, y]$ can be written $r = \sum_{m \in B} a_m(z) m + L_x P + L_y Q$ with $a_m \in k[z^{-1}]$ and $P, Q \in z^{-1} k[z^{-1}][x, y]$.*
- (b) *If $\sum_{m \in B} a_m m = L_x P + L_y Q$ with $a_m \in F_0$ and $P, Q \in F_0[x, y]$, then all $a_m = 0$.*
- (c) *Consequently B is an F_0 -basis of H_q^2 and a $k[z, z^{-1}]$ -basis of $G = \Omega^2[z, z^{-1}]/d_z \Omega^1[z, z^{-1}]$. In this basis $\nabla_z[m] = [q m] = \sum_{m'} A_{mm'}[m']$ with*

$$A = A_0 + A_1 z^{-1} + \cdots + A_s z^{-s} \in M_\mu(k[z^{-1}]),$$

and A_0 is the matrix of multiplication by q on $k[x, y]/(q_x, q_y)$ in the basis B . Under (M) its eigenvalues are $c_1, \dots, c_\mu$, each simple.

The same statements hold with k replaced by $\mathbb{C}$, and the reduction and independence arguments also apply after specialization at every $z = \xi \neq 0$.

Proof. (a) Let $x^m y^n$ be a monomial of r , with coefficient γ , of maximal weight among those not in B . If $m \geq a - 1$ put $v = \gamma x^{m-a+1} y^n / (a\alpha z)$. Then $L_x v = \gamma x^m y^n + (\gamma/a\alpha) q_{<,x} x^{m-a+1} y^n + v_x$, and the last two terms have weight strictly below $\text{wt}(x^m y^n)$ because $\text{wt}(q_{<,x}) < ab - b$. Replace r by $r - L_x v$. If $m < a - 1$ then $n \geq b - 1$ and one uses L_y symmetrically. The maximal weight of a monomial outside B , with its multiplicity among the finitely many monomials of that weight, strictly decreases, so the process stops.

(b) Among all pairs (P, Q) representing the given combination choose one minimizing $\varpi = \max(\text{wt } P + ab - b, \text{wt } Q + ab - a)$. Let P^{top} and Q^{top} be the components of P and Q of weights $\varpi - ab + b$ and $\varpi - ab + a$ (one of them may be zero). The weight- ϖ component of $L_x P + L_y Q$ is $z(a\alpha x^{a-1} P^{\text{top}} + b\beta y^{b-1} Q^{\text{top}})$. If it is nonzero it is a nonzero element of the monomial ideal (x^{a-1}, y^{b-1}) , which cannot equal the weight- ϖ component of a combination of monomials of B . Hence it vanishes, and since x^{a-1}, y^{b-1} is a regular sequence, $P^{\text{top}} = b\beta y^{b-1} R$, $Q^{\text{top}} = -a\alpha x^{a-1} R$. As L_x and L_y commute, $(P - L_y(R/z), Q + L_x(R/z))$ represents the same element and has smaller ϖ unless $P = Q = 0$. So $P = Q = 0$ and all $a_m = 0$.

(c) The first two assertions follow from (a) and (b). In (a), the terms of order z^0 give $r \equiv \sum a_m(\infty) m$ modulo (q_x, q_y) , and the argument of (b) with z removed shows B is a basis of the Jacobian algebra; this identifies A_0 . That algebra is the product of its local algebras at the critical points, on each of which q has the single eigenvalue $q(P)$, with multiplicity the local Milnor number. $\square$

6 Period calculus and irreducibility of H_q^2

Let $S = \mathbb{C} \setminus \{c_1, \dots, c_\mu\}$, $C_t = q^{-1}(t)$. The arguments below use three geometric facts.

- (G1) q is a locally trivial fibration over S ; every fibre of q is reduced and irreducible; for $t \in S$, $H_1(C_t; \mathbb{C})$ has dimension μ and its intersection form is nondegenerate.
- (G2) The local monodromy T_i around c_i is the Picard–Lefschetz transvection $\delta \mapsto \delta \pm \langle \delta, \nu_i \rangle \nu_i$, and the vanishing period $\int_{\nu_i(t)} dx \wedge dy/dq$ is holomorphic at c_i with a nonzero value there; in particular $\nu_i \neq 0$ and $T_i \neq 1$.
- (G3) $\pi_1(S)$ acts irreducibly on $H_1(C_t; \mathbb{C})$.

Proposition 6.1 (Geometry of the class (W)). *For q of class (W) satisfying (M), (G1)–(G3) hold.*

Proof. Fibres. The top weighted form of $q - t$ is $\alpha x^a + \beta y^b$, which is irreducible in $\mathbb{C}[x, y]$ because $\gcd(a, b) = 1$. The top form of a product is the product of the top forms, so $q - t$ is irreducible; in particular every fibre is reduced and irreducible.

Tameness. Put $R = \max(|x|^{1/b}, |y|^{1/a})$, $X = x/R^b$, $Y = y/R^a$. Then $R^{-(ab-b)} q_x = a\alpha X^{a-1} + o(1)$ and $R^{-(ab-a)} q_y = b\beta Y^{b-1} + o(1)$ uniformly on $\max(|X|, |Y|) = 1$ as $R \rightarrow \infty$, and one of the two leading terms has modulus bounded below there. Hence $\|\nabla q\| \rightarrow \infty$ at infinity, so q is tame in the sense of Broughton [4, Definition 3.1]: $\|\partial q\|$ is bounded away from 0 outside a compact

neighbourhood of the critical points. For tame polynomials, [4, p. 230] shows that q is locally trivial over any region of the target containing no critical value, and [4, Theorem 1.2 and p. 231] gives $H_1(C_t; \mathbb{Z}) \cong \mathbb{Z}^{\mu'}$ for a regular value t , where μ' is the sum of the Milnor numbers at the critical points; by Lemma 5.1(c), $\mu' = \dim \mathbb{C}[x, y]/(q_x, q_y) = \mu$.

Local model and stability. Write

$$f_\epsilon(X, Y) = \alpha X^a + \beta Y^b + \sum_{bi+aj < ab} q_{ij} \epsilon^{ab-bi-aj} X^i Y^j.$$

Every exponent of ϵ is a positive integer, so this is a holomorphic deformation of $f_0 = \alpha X^a + \beta Y^b$. For $\epsilon \neq 0$ it equals $\epsilon^{ab} q(\epsilon^{-b} X, \epsilon^{-a} Y)$; its critical points and values are $(\epsilon^b x_i, \epsilon^a y_i)$ and $\epsilon^{ab} c_i$. Choose a Milnor ball $\overline{B}_r$ for f_0 , and choose $\eta > 0$ so that its levels $|t| \leq 2\eta$ are transverse to ∂B_r . Compactness and C^1 convergence preserve this transversality for f_ϵ with $|t| \leq \eta$ when $|\epsilon|$ is sufficiently small. Shrink the parameter disk so that all critical points lie in $B_{r/2}$ and all critical values have modulus less than $\eta/2$.

At the fixed value $t_* = \eta$, the family

$$\{(\epsilon, X, Y) : f_\epsilon(X, Y) = t_*, (X, Y) \in \overline{B}_r\} \longrightarrow \{\epsilon : |\epsilon| < \epsilon_0\}$$

is a proper submersion, including on its boundary. The boundary form of Ehresmann's theorem therefore identifies its fibres with the Milnor fibre of f_0 . This is also the perturbation comparison in [1, §2.1, pp. 29–31]. For a fixed small nonzero ϵ , the local proper submersion over the disk minus the critical values transports this identification to every regular level there. Put $S_t = f_\epsilon^{-1}(t) \cap \overline{B}_r$ and $C_t = f_\epsilon^{-1}(t)$. By [1, Theorem 2.1, p. 31] a distinguished set of vanishing cycles $\Delta_1, \dots, \Delta_\mu$ is a basis of $H_1(S_t; \mathbb{Z})$. For a plane-curve singularity the kernel of the intersection form has dimension $r_0 - 1$, where r_0 is the number of branches [1, §3.6, p. 100]. The binomial germ f_0 is irreducible, so this form is nondegenerate; compare also [1, p. 411, Example 2].

Compatible local and global transport. Tameness is preserved by the fixed nonzero rescaling. Over a relatively compact target neighborhood U whose closure avoids the critical values, $\|\nabla f_\epsilon\|$ is bounded below: use tameness outside a compact set and compactness inside it. A bounded horizontal lift of a constant target tangent vector $v \in \mathbb{C}$ is

$$V_v = v \frac{\overline{\nabla f_\epsilon}}{\|\nabla f_\epsilon\|^2}, \quad df_\epsilon(V_v) = v.$$

Boundary transversality supplies a smooth horizontal lift tangent to ∂B_r . On a compact collar patch it to V_v by a partition of unity. The patched lift remains horizontal and bounded and is tangent to the sphere. Its flows exist for the required finite times, preserve the sphere and its two sides, and yield compatible local trivializations of the pair (C_t, S_t) . This uses both tameness and transversality, following the bounded-lift construction in [4, pp. 229–230].

Inclusion $j : S_t \hookrightarrow C_t$ preserves intersection numbers. Hence $j_* v = 0$ implies $\langle v, u \rangle = \langle j_* v, j_* u \rangle = 0$ for every u , and local nondegeneracy gives $v = 0$. The global and local homology groups both have dimension μ , so j_* is an isomorphism. The compatible transport above makes it an isomorphism of local systems. The punctured disk contains all critical values and its inclusion in their complement in $\mathbb{C}$ induces an isomorphism on fundamental groups. Thus the global intersection form is nondegenerate and its monodromy is that of this morsification. Rescaling back to q proves the remaining assertions of (G1).

(G2). The Picard–Lefschetz formula is [1, §1.3, p. 26]. At a nondegenerate critical point choose holomorphic coordinates with $q = c_i + uv$, $dx \wedge dy = J du \wedge dv$. The vanishing cycle is $\{uv =$

$t - c_i$, $|u| = r\}$ and $dx \wedge dy/dq = -J du/u$ on C_t , so the vanishing period is $\mp 2\pi i \sum_{\kappa} J_{\kappa\kappa}(t - c_i)^{\kappa}$, holomorphic with value $\mp 2\pi i J(0) \neq 0$ at c_i . Hence $\nu_i \neq 0$, and $T_i \neq 1$ by nondegeneracy of the form.

(G3). By the previous steps $H_1(C_t)$ has the basis $\Delta_1, \dots, \Delta_{\mu}$, the global monodromy group is generated by the Picard–Lefschetz transformations T_{Δ_i} , and the form is nondegenerate. By the theorem of Gabriellov and Lazzeri [1, Theorem 3.3], the diagram of a singularity relative to any distinguished basis is connected: the graph on $\{1, \dots, \mu\}$ with an edge when $\langle \Delta_i, \Delta_j \rangle \neq 0$ is connected. (The diagram is defined after stabilization, which changes off-diagonal intersection numbers only by signs; self-intersections may change.) Let $W \neq 0$ be invariant. By nondegeneracy there are $w \in W$ and i with $\langle w, \Delta_i \rangle \neq 0$; then $(T_{\Delta_i} - 1)w = \pm \langle w, \Delta_i \rangle \Delta_i$ puts Δ_i in W , and $(T_{\Delta_j} - 1)\Delta_i = \pm \langle \Delta_i, \Delta_j \rangle \Delta_j$ propagates membership along the connected graph. So $W = H_1$. $\square$

For a horizontal family of cycles $\delta(t) \subset C_t$ over a simply connected $U \subset S$ and $\omega = \sum_m z^m \omega_m \in \Omega_{\mathbb{C}[x,y]}^2[z]$ put

$$\pi_{\omega_m}(t) = \int_{\delta(t)} \frac{\omega_m}{dq}, \quad \Pi_{\delta}(\omega)(t) = \sum_m (-\partial_t)^m \pi_{\omega_m}(t).$$

Lemma 6.2 (Weyl intertwining). $\Pi_{\delta}(z\omega) = -\partial_t \Pi_{\delta}(\omega)$, $\Pi_{\delta}(\partial_z \omega + q\omega) = t \Pi_{\delta}(\omega)$, and $\Pi_{\delta}(d_z \beta) = 0$ for $\beta \in \Omega^1[z]$.

Proof. The first is the definition. Since $q = t$ on C_t , $\pi_{q\omega_m} = t\pi_{\omega_m}$, and $(-\partial_t)^m t = t(-\partial_t)^m - m(-\partial_t)^{m-1}$ gives the second. For a polynomial one-form β_0 , $\pi_{dq \wedge \beta_0} = \int_{\delta(t)} \beta_0$ and $\pi_{d\beta_0} = \partial_t \int_{\delta(t)} \beta_0$ (Gelfand–Leray); the two contributions to $\Pi_{\delta}((d + z dq \wedge)\beta)$ cancel termwise. $\square$

Lemma 6.3 (Period injectivity). Assume (G1), (G3) and $\mu \geq 2$. If $\Pi_{\delta}(\omega) \equiv 0$ for every δ , then $[\omega] = 0$ in $H_q^2 \otimes \mathbb{C}$.

Proof. If $d\theta = \omega'$ then $d_z \theta = \omega' + z dq \wedge \theta$, so $[\omega'] = -z[dq \wedge \theta]$. Write $\omega = \sum_{m=0}^N z^m \omega_m$ and apply this to the lowest coefficient: subtracting $d_z \theta_0$ removes the z^0 term and changes only the coefficient of z^1 . After N steps $\omega \equiv z^N \omega^{\#}$ modulo $d_z \Omega^1[z]$ with $\omega^{\#}$ independent of z , and $\Pi_{\delta}(\omega) = (-\partial_t)^N \pi_{\omega^{\#}}$ by Lemma 6.2. Thus every branch of every period of $\alpha^{\#} = \omega^{\#}/dq$ is a polynomial of degree $< N$ (zero when $N = 0$), hence single valued: $\int_{(g-1)\delta} \alpha^{\#} = 0$ for all $g \in \pi_1(S)$. The span of the $(g-1)\delta$ is an invariant subspace, nonzero because an irreducible representation of dimension ≥ 2 is nontrivial; by (G3) it is all of H_1 . So all periods of $\alpha^{\#}$ vanish. Choose $\theta^{\#}$ with $d\theta^{\#} = \omega^{\#}$. Its periods have derivative zero, hence are constant, hence single valued, hence (same argument) zero. So $\theta^{\#}$ is exact along the generic fibres. The critical locus is finite and all fibres are connected and nonempty; these give Bonnet’s primitive and quasi-fibered hypotheses for a map to the line. Thus [3, Definitions 1.3–1.4 and Theorem 1.5] gives $\theta^{\#} = dR + a dq$. Then $\omega^{\#} = da \wedge dq = d_z(a dq)$ and $[\omega] = \pm z^N [\omega^{\#}] = 0$. $\square$

Proposition 6.4 (Irreducibility). Assume (G1)–(G3) and $\mu \geq 2$. Every nonzero element of $H_q^2 \otimes_{F_0} F$ has minimal differential operator of order $\geq \mu$ over F . With Lemma 5.1, $H_q^2 \otimes F$ is an irreducible differential module of rank μ .

Proof. Let $\eta \neq 0$. Multiplying η by an element of $F^{\times}$ changes neither the differential submodule it generates nor the minimal order of an annihilator (Leibniz), so we may assume $\eta = [\omega]$ with $\omega \in \Omega_{\mathbb{C}}^2[z]$. Let $P = \sum_{l=0}^r p_l(z) \nabla_z^l$, $p_l \in \mathbb{C}[z]$ not all zero, satisfy $P\eta = 0$. Multiplying P on the left

by a polynomial in z , which does not change r , we may assume $P(z, \nabla_z)\omega = d_z\beta$ with $\beta \in \Omega^1[z]$. By Lemma 6.2, for every δ ,

$$\widehat{P}\Pi_\delta(\omega) = 0, \quad \widehat{P} = \sum_l p_l(-\partial_t)t^l = \sum_{n=0}^N a_n(t)\partial_t^n, \quad a_N(t) = (-1)^N \sum_l p_{l,N}t^l,$$

where $N = \max \deg p_l$ and $p_{l,N}$ is the coefficient of z^N in p_l . So $a_N \neq 0$ and $\deg a_N \leq r$. The map $\delta \mapsto \Pi_\delta(\omega)$ from $H_1(C_{t_0}; \mathbb{C})$ to germs at t_0 is linear, compatible with analytic continuation, and nonzero by Lemma 6.3; by (G3) it is injective. If $a_N(c_i) \neq 0$, every solution germ of $\widehat{P}$ near c_i is holomorphic at c_i , so continuation around c_i is trivial on the image, contradicting $T_i \neq 1$. Hence a_N vanishes at $c_1, \dots, c_\mu$ and $r \geq \deg a_N \geq \mu$. A proper nonzero submodule would contain an element with minimal operator of order $< \mu$. $\square$

7 The exponential torus and SL_μ

Consider $Y' = AY$ with A as in Lemma 5.1 and let $G \subset \mathrm{GL}_\mu$ be its differential Galois group over F , acting on the solution space V .

Lemma 7.1 (Formal diagonalization). *Let $A = \sum_{j \geq 0} A_j w^j$, $w = 1/z$, with A_0 having distinct eigenvalues $c_1, \dots, c_n$. Then $Y' = AY$ has a formal fundamental matrix $\widehat{Y} = \widehat{H}(w)z^\Lambda e^{Cz}$ with $C = \mathrm{diag}(c_i)$, Λ a constant diagonal matrix and $\widehat{H} \in \mathrm{GL}_n(\mathbb{C}[[w]])$.*

Proof. After a constant change of basis $A_0 = C$. Seek $\widehat{H} = \sum_{k \geq 0} H_k w^k$, $H_0 = I$, with $\widehat{H}' = A\widehat{H} - \widehat{H}(C + \Lambda w)$; then $\widehat{H}z^\Lambda e^{Cz}$ is a solution. Since $(w^k)' = -kw^{k+1}$, the coefficient of w^k gives, for $k \geq 1$,

$$-(k-1)H_{k-1} = [C, H_k] + \sum_{j=1}^k A_j H_{k-j} - H_{k-1}\Lambda.$$

For $k = 1$ put $\Lambda = \mathrm{diag}(A_1)$; the off-diagonal entries $(c_i - c_l)(H_1)_{il}$ of $[C, H_1]$ determine the off-diagonal part of H_1 . For $k \geq 2$ the diagonal part of the equation reads $-(k-1)\mathrm{diag} H_{k-1} = \mathrm{diag}(A_1^{\mathrm{off}} H_{k-1}^{\mathrm{off}} + \sum_{j=2}^k A_j H_{k-j})$, which determines $\mathrm{diag} H_{k-1}$ from quantities already known, and the off-diagonal part determines H_k^{off} . Induction on k constructs $\widehat{H}$. $\square$

Proposition 7.2 (Separated-root group criterion). *Let $A \in M_\mu(\mathbb{C}[z^{-1}])$, $\mu \geq 2$, define an irreducible differential module. Suppose its leading matrix A_0 has simple eigenvalues $c_1, \dots, c_\mu$ satisfying (R). Then its differential Galois group over F contains SL_μ . In particular this holds for the class (W) under (M), (R), by Propositions 6.1 and 6.4.*

Proof. Formal solution and a diagonal subgroup. Let $\mathfrak{K} = \mathbb{C}((w))$ with $' = d/dz = -w^2 d/dw$. By the hypothesis on A and Lemma 7.1, $\widehat{H}' = A\widehat{H} - \widehat{H}(C + \Lambda w)$ with $C = \mathrm{diag}(c_i)$, $\Lambda = \mathrm{diag}(\lambda_i)$, $\widehat{H} \in \mathrm{GL}_\mu(\mathbb{C}[[w]])$. Let K_0 be a Picard–Vessiot extension of $\mathfrak{K}$ for the diagonal system $Y' = (C + \Lambda w)Y$; it is generated by nonzero u_i with $u_i' = (c_i + \lambda_i w)u_i$. Then $\widehat{Y} = \widehat{H} \mathrm{diag}(u_i)$ is a fundamental matrix of $Y' = AY$ in K_0 , and since K_0 has constants $\mathbb{C}$, $\widehat{L} = F(\widehat{Y}_{ij})$ is a Picard–Vessiot extension of F ; G is its group, written in the basis of columns of $\widehat{Y}$. Every $\sigma \in \Gamma = \mathrm{Gal}(K_0/\mathfrak{K})$ multiplies each u_i by a constant, so it restricts to an element of G which is diagonal. Let $D_\Gamma \subset G$ be the resulting closed subgroup of the diagonal torus. A character $x^n = \prod x_i^{n_i}$ is trivial on D_Γ iff $\prod u_i^{n_i}$ is Γ -fixed, i.e. lies in $\mathfrak{K}$. An element $f \in \mathfrak{K}^\times$ has $f'/f \in w\mathbb{C}[[w]]$, while $(\prod u_i^{n_i})'/\prod u_i^{n_i} = \sum n_i c_i + (\sum n_i \lambda_i)w$;

so such n satisfy $\sum n_i c_i = 0$. A closed subgroup of a torus is the common kernel of the characters trivial on it, hence

$$D_\Gamma \supseteq T := \{d : d^n = 1 \text{ whenever } \sum n_i c_i = 0\}.$$

T is a torus with character group $\mathbb{Z}^\mu / \{n : \sum n_i c_i = 0\} \cong \sum \mathbb{Z} c_i$, acting on the i th basis vector with weight c_i ; being connected, $T \subset G^0$.

Root decomposition. By (R) the weights $c_i - c_j$ ($i \neq j$) of T on $\mathfrak{gl}_\mu$ are pairwise distinct and nonzero, so $\mathfrak{g} = \text{Lie } G^0 = (\mathfrak{g} \cap \mathfrak{d}) \oplus \bigoplus_{(i,j) \in R} \mathbb{C} E_{ij}$, $\mathfrak{d}$ the diagonal matrices. G is reductive because V is a faithful irreducible representation by hypothesis, so G^0 is reductive; $\mathfrak{g} \cap \mathfrak{d}$ is the centralizer of $\text{Lie } T$, a Cartan subalgebra, and therefore $R = -R$. Also R is closed under $(i, j), (j, l) \mapsto (i, l)$. So R defines an equivalence relation with classes $I_1, \dots, I_s$; each $V_{I_a} = \text{span}\{e_i : i \in I_a\}$ is an irreducible G^0 -module and they are pairwise nonisomorphic, so the G^0 -submodules of V are exactly the sums of the V_{I_a} .

One class. Since G^0 is normal, G permutes the V_{I_a} . The fixed field E_0 of G^0 in $\widehat{L}$ is a finite Galois extension of F with group G/G^0 . Under the isomorphism of $\widehat{L}$ with the Picard–Vessiot field generated by an analytic fundamental matrix, its elements are algebraic functions meromorphic on the universal cover of $\mathbb{C}^\times$ (the system is singular only at $0, \infty$), so $E_0 = F(\zeta)$ with $\zeta^d = z$. The image of D_Γ in $\text{Gal}(E_0/F)$ has fixed field $E_0 \cap \mathfrak{K}$. This is $F(\zeta^{d/d'})$ for some $d' \mid d$, and $(\zeta^{d/d'})^{d'} = w^{-1}$ forces $d' = 1$ because w -orders in $\mathfrak{K}$ are integers. So D_Γ maps onto G/G^0 , i.e. $G = G^0 D_\Gamma$. Diagonal matrices preserve each V_{I_a} , so each V_{I_a} is G -stable and irreducibility gives $s = 1$. Then every E_{ij} lies in $\mathfrak{g}$ and $\text{SL}_\mu \subset G^0$. $\square$

8 A nonempty open phase locus

Proposition 8.1 (Zariski-open separated critical values). *Fix coprime $a, b \geq 2$ and nonzero leading coefficients α, β . In the affine space of all coefficients of $q_<$ of weight less than ab , the conditions $(M), (R)$ define a nonempty Zariski-open subset.*

Proof. The leading partial derivatives form the monomial regular sequence x^{a-1}, y^{b-1} . The weighted division used in Lemma 5.1 therefore gives a finite free universal Jacobian algebra of rank μ . Its reduced-fibre locus, with distinct critical values, is open. It is nonempty: take univariate Morse polynomials in the two variables and choose their lower coefficients so that the two finite sets of critical values have distinct pairwise sums. Scaling the critical points of one univariate polynomial varies its critical values by a nonzero scalar while keeping its prescribed leading coefficient; only finitely many scalars are excluded. Call this irreducible open parameter space U .

Over U the critical-point scheme is finite etale. Pass to its finite etale cover labeling all points P_i and values c_i . For an allowed coefficient variation h , differentiation of $c_i = q(P_i)$ gives $dc_i(h) = h(P_i)$, since $dq(P_i) = 0$.

Suppose first $a, b \geq 3$. Every polynomial of ordinary degree at most two has weight less than ab . At a forbidden equality $c_i + c_l - c_j - c_k = 0$, its differential cannot vanish on all these variations. Indeed equality of evaluations for every linear and quadratic h says that the pairs (P_i, P_l) and (P_j, P_k) have equal sums and second-moment matrices. Writing a pair as $(s + v)/2, (s - v)/2$, those data determine vv^T ; over a field of characteristic zero, $vv^T = ww^T$ forces $w = \pm v$. Thus the unordered pairs coincide, including multiplicities. Under $i \neq j, k \neq l$, the only possibility is the identical ordered difference $(i, j) = (k, l)$, which was excluded. Each forbidden equality therefore defines a proper closed subset on every component of the cover.

If $b = 2$, completing the square in $q = \beta y^2 + P(x)y + Q(x)$ gives a degree- a polynomial in x . On U its $a - 1$ critical x -coordinates are distinct. The allowed variations $1, x, \dots, x^{a-2}$ have an

invertible Vandermonde evaluation matrix, so no forbidden equality has identically zero differential. The case $a = 2$ is symmetric. The finite union of the forbidden loci has proper closed image under the finite label cover. Its complement in U is the required nonempty open subset. $\square$

Example 8.2 (Strictly weaker than rational independence). Let

$$p = x^5 - \frac{5}{4}x^4 - \frac{35}{3}x^3 + \frac{5}{2}x^2 + 30x, \quad q = p(x) + y^2.$$

Then $p' = 5(x+2)(x+1)(x-1)(x-3)$ and its critical values are

$$-\frac{26}{3}, \quad -\frac{217}{12}, \quad \frac{247}{12}, \quad -\frac{243}{4}.$$

Their twelve ordered differences are distinct (an exact rational check), so (M),(R) hold. The values are rational, so the three differences from one base value are rationally dependent: (Q) fails.

Remark 8.3. The proposition concerns the full weighted coefficient space, not every sparse subfamily. A separable family with two critical points in each variable has additive rectangles and can fail (R) identically. Nor are (S),(S*),(E) Zariski-open conditions on that space: they concern an actual real chain. Root separation is a sufficient group certificate; its failure does not imply that the Galois group is smaller.

9 Face disjointness

Stable lattices at infinity. Put $R_\infty = \mathbb{C}[[w]]$, $K_\infty = \mathbb{C}((w))$ and $\delta = d/dz = -w^2 d/dw$. A lattice in a differential module over K_∞ is a finite free R_∞ -submodule spanning it. A lattice L is stable if $\partial L \subset L$. Since $\delta(R_\infty) \subset wR_\infty$, its operator induces a $\mathbb{C}$ -linear operator on L/wL , called the leading operator of this stable lattice. In the coefficient-module convention its matrix for a system $Y' = E(w)Y$ is $E(0)^\top$, which has the same spectrum as $E(0)$.

Lemma 9.1 (Stable lattices and subquotients). *Let $L \subset M$ be a stable lattice and $N \subset M$ a differential submodule. Then $L_N = L \cap N$ and $L_{M/N} = \text{im}(L \rightarrow M/N)$ are stable lattices, and*

$$0 \longrightarrow L_N/wL_N \longrightarrow L/wL \longrightarrow L_{M/N}/wL_{M/N} \longrightarrow 0$$

is exact and compatible with the leading operators. Their characteristic polynomials therefore multiply. In particular a direct sum of successive quotients of a differential-module filtration has a stable lattice with the same leading characteristic polynomial as L .

Proof. The intersection is a saturated, finitely generated submodule of L : if $wv \in L \cap N$ and $v \in L$, then $v \in N$ because N is a K_∞ -vector space. The intersection and the quotient are free over the discrete valuation ring R_∞ and span the desired spaces. Stability is inherited. Since the quotient is free, reduction modulo w preserves exactness. Also $\delta(w) = -w^2$, so each leading operator is well-defined and the sequence intertwines them. Apply this successively to the filtration. $\square$

Lemma 9.2 (Detection of a rank-one leading coefficient). *If a module M with stable lattice L contains a nonzero vector v with $\partial v = (d + \rho w)v$, $d, \rho \in \mathbb{C}$, then d is an eigenvalue of the leading operator on L/wL .*

Proof. The saturated lattice $L \cap K_\infty v$ has a generator fv , where $f = w^m u$ with $u \in R_\infty^\times$. Its coefficient is

$$d + \rho w + \frac{\delta f}{f}, \quad \frac{\delta f}{f} = -mw - w^2 \frac{u_w}{u} \in wR_\infty.$$

Its leading eigenvalue is therefore d . Its reduction injects into L/wL by Lemma 9.1. $\square$

Corollary 9.3 (A leading-spectrum obstruction). *Suppose M has a stable lattice with leading spectrum Σ . Let P admit a basis over K_∞ with $\partial e_i = (c_i + \lambda_i w)e_i$. If*

$$\text{End}^0(P) \hookrightarrow \text{End}(M^{\text{ss}})$$

is a differential-module embedding, then $c_i - c_j \in \Sigma - \Sigma$ for all $i \neq j$.

Proof. By Lemma 9.1, M^{ss} has a stable lattice L^{ss} with the same leading characteristic polynomial. The lattice $\text{End}_{R_\infty}(L^{\text{ss}})$ is stable; its leading operator is the commutator with that of L^{ss} , with eigenvalues $s - s'$ for $s, s' \in \Sigma$. The vector $e_i \otimes e_j^*$ is trace-free for $i \neq j$ and has differential coefficient $c_i - c_j + (\lambda_i - \lambda_j)w$. Its image in the endomorphism module is nonzero, so Lemma 9.2 gives the result.

If M^{ss} was taken over F first, extend a composition series from F to K_∞ and apply the lattice lemma to this filtration. Its quotients need not remain simple. No assertion that semisimplification commutes with this extension is used. $\square$

For a face polynomial p of degree $d \geq 2$, the one-variable reduction gives a matrix in $M_{d-1}(k[z^{-1}])$ whose leading matrix is multiplication by p on $k[s]/(p')$. Thus its stable lattice has as leading spectrum the critical values of p , also when those values repeat. For the interior module, Lemma 7.1 supplies the diagonal basis required in Corollary 9.3. These are all the local facts needed below; no general formal-normal-form theorem is required.

Let $\mathcal{B}$ be the closed homogeneous system satisfied by the boundary vector (reduced face moments, endpoint exponentials, 1), $\mathcal{T}_\partial$ its tensor category and G_∂ its Galois group.

Proposition 9.4. *Suppose $G \supseteq \text{SL}_\mu$, $\mu \geq 2$, and that the interior module has the formal diagonalization of Lemma 7.1. If $\text{End}^0(A)$ belongs to $\mathcal{T}_\partial$, then there is a face e with $d_e \geq 2$ such that every $c_i - c_j$ is a difference of two critical values of p_e . In particular (F) implies $\text{End}^0(A) \notin \mathcal{T}_\partial$.*

Proof. The Galois group of $\text{End}^0(A)$ is PGL_μ , so there is a surjection $\rho : G_\partial \rightarrow \text{PGL}_\mu$ through which G_∂ acts on $\mathfrak{sl}_\mu$ by $\text{Ad} \circ \rho$. The unipotent radical dies, so ρ factors through $G' = \text{Gal}(\mathcal{B}^{\text{ss}}) \subset \prod_e \text{GL}(W_e) \times (\text{torus})$, W_e the solution space of $H_{p_e}^{\text{ss}}$. Write $\text{Lie } G' = \mathfrak{z} \oplus \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_r$. Since $\mathfrak{sl}_\mu$ is simple, $\ker d\rho$ contains $\mathfrak{z}$ and all simple ideals but one, $\mathfrak{s}$; as $\ker d\rho$ is $\text{Ad}(G')$ -stable, so is $\mathfrak{s}$, and $d\rho : \mathfrak{s} \rightarrow \mathfrak{sl}_\mu$ is a G' -equivariant isomorphism. The projections $\mathfrak{s} \rightarrow \mathfrak{gl}(W_e)$ are G' -equivariant and each is injective or zero; they are not all zero because the remaining factor is abelian. For an e with injective projection, the G_∂ -module $\mathfrak{sl}_\mu$ embeds in $\text{End}(W_e)$. Using the covariant horizontal-fibre equivalence and the trace dualities noted above, this gives a differential-module embedding

$$\text{End}^0(A) \hookrightarrow \text{End}(H_{p_e}^{\text{ss}}).$$

For the face module the leading spectrum is the set of critical values of p_e . Corollary 9.3, after extension to K_∞ , therefore makes every $c_i - c_j$ a difference of two such critical values. This contradicts (F). $\square$

Corollary 9.5 (Intrinsic, spectral, and rank tests). *The full comparison needs the intrinsic exclusion $\text{End}^0(A) \notin \mathcal{T}_\partial$. Under the hypotheses of Proposition 9.4, either (F) or the inequalities*

$$d_e - 1 < \mu \quad \text{for every face with } d_e \geq 2$$

certifies this exclusion. The rank test is automatic for straight faces in class (W) when $\min(a, b) \geq 3$.

Proof. The proof of Proposition 9.4 produces an embedding into $\text{End}(H_{p_e}^{\text{ss}})$ for one single face if the exclusion fails. Its dimensions would require $\mu^2 - 1 \leq (d_e - 1)^2$, impossible when $d_e - 1 < \mu$. For straight faces $d_e \leq \max(a, b)$ and $\max(a, b) - 1 < (a - 1)(b - 1)$. No comparison merely with the dimension of a boundary generating vector would control its whole tensor category; the single-face reduction is indispensable. $\square$

10 The saddle lemma

Let $\mathcal{N} = \text{span}_F\{1, \text{reduced face moments, endpoint exponentials}\}$.

Lemma 10.1. *Assume (G1)–(G3), $\mu \geq 2$, and (S) or (S*). Then $I_D(dx \wedge dy) \notin \mathcal{N}$.*

Proof. Suppose $P(z)I_D(dx \wedge dy) = \sum_j R_j(z)g_j$ with $0 \neq P$, $R_j \in \mathbb{C}[z]$ and g_j among the generators. By Lemma 1.4(b), the inverse Laplace transforms of the face generators have algebraic densities on intervals free of the finitely many face-critical, vertex and zero values; endpoint exponentials and 1 give point masses. The compactly supported distributions are determined by their Laplace transforms: restrict an identically zero entire transform to the imaginary axis and apply Fourier uniqueness. Hence on such an interval the analytic density $\rho = \rho_{D,1}$ of $I_D(dx \wedge dy)$ satisfies: $g = P(-\partial_t)\rho$ is algebraic over $\mathbb{C}(t)$. Under (S) this applies to $(c, c + \varepsilon)$ for small $\varepsilon > 0$.

Local form under (S). By the real-analytic Morse lemma there are local coordinates with $q = c + uv$, $dx \wedge dy = J du \wedge dv$, $J(0)^2 = |\det \text{Hess } q(P_0)|^{-1}$. Take a box $\{|u|, |v| \leq \delta\}$ on which the multiplicity of D is w . After shrinking δ the edges of the box are transverse to the levels near c and its corners have levels $c \pm \delta^2 \neq c$. Outside the open box the level curves near level c meet neither a critical point (distinct critical values) nor a tangency or vertex of ∂D (by (S)), so that contribution to ρ is analytic at c . Inside, with $s = t - c$,

$$\rho_{\text{box}}(s) = \pm w \int_{|s|/\delta \leq |u| \leq \delta} J(u, s/u) \frac{du}{|u|} = \pm w(-2j(s) \log s + \text{analytic}) \quad (0 < s < \delta^2),$$

where $J = \sum J_{\kappa\lambda} u^\kappa v^\lambda$ and $j(s) = \sum_\kappa J_{\kappa\kappa} s^\kappa = \frac{1}{2\pi i} \oint_{|u|=r} J(u, s/u) \frac{du}{u}$; the terms with $\kappa \neq \lambda$ integrate to convergent power series in s . Since $dx \wedge dy/dq = -J du/u$ on C_t , j is $\mp \frac{1}{2\pi i}$ times the period $h(t) = \int_{\nu(t)} dx \wedge dy/dq$ over the vanishing cycle, and $j(0) = J(0) \neq 0$. Thus ρ continues around c with variation a nonzero constant times h , which is (S*).

Conclusion from (S).* Write the variation as λh , $\lambda \neq 0$. The vanishing period h is fixed by the local monodromy, so N turns around c_i change ρ by $N\lambda h$. The germ g continues with ρ and remains algebraic, so for some $N \geq 1$ it returns to itself; therefore $N\lambda P(-\partial_t)h = 0$ near c_i and h extends to an entire function. So h is single valued on S , i.e. $\int_{(g-1)\nu_i} dx \wedge dy/dq = 0$ for all $g \in \pi_1(S)$. As in Lemma 6.3 the $(g-1)\nu_i$ span H_1 ($\nu_i \neq 0$ is not invariant, by (G3) and $\mu \geq 2$), so the period over ν_i itself vanishes, contradicting (G2). $\square$

11 Proof of the linear theorem

For (W),(M), Propositions 6.1 and 6.4 give irreducibility of $H_q^2 \otimes F$, independently of (R),(Q),(F). Lemma 10.1 supplies nonzero detection under (S) or (S*); Lemma 3.2 supplies it under (E). Apply Proposition 3.1. The combined system has no finite pole except zero by Lemma 5.1 and the one-variable boundary reduction, so the same proposition proves the value assertion for every $\xi \in k^\times$.

Specializing the Laurent Stokes identity is legitimate at every such ξ . Its boundary term is in $\mathcal{V}_{\partial,\xi}$, so linear comparison forces all remainder coefficients to vanish if the integral lies in that span. The specialized independence part of Lemma 5.1 identifies this with polynomial twisted exactness. Conversely exactness gives membership by Stokes. If $r \in \text{span}_k B$ is nonzero, its remainder is the same nonzero constant vector at every ξ ; since $k \subset \mathcal{V}_{\partial,\xi}$, its integral is transcendental over k . This proves Theorem 1.5.

12 Proof of Theorem 1.7

(i) is Lemma 5.1. For (ii): integrating the certificates of qm gives $M' = AM + b$ with $b_m = \int_{\partial D} e^{zq} (P_{qm} dy - Q_{qm} dx) \in \mathcal{N}$. The actual one-column criterion, Theorem 4.3, requires: (1) $G \supseteq \text{SL}_\mu$, which is Proposition 7.2; (2) $\text{End}^0(A) \notin \mathcal{T}_\partial$, which is the assumed intrinsic exclusion or follows from Corollary 9.5; (3) some $M_m \notin \mathcal{N}$, which is Lemma 10.1 or Lemma 3.2, since $1 \in B$. Hence the M_m are algebraically independent over $F(\mathcal{N}) = K_\partial$.

(iii) The vector (boundary coordinates, M) consists of E -functions (Theorem 2.1) and satisfies a homogeneous linear system over $k(z)$ whose only finite pole is $z = 0$: for M by Lemma 5.1, for the faces by its one-variable case. Let d be the transcendence degree of $K_{\partial,0}$ over $k(z)$. Since $K_{\partial,0} \subset K_\partial = \mathbb{C}K_{\partial,0}$, the algebraic independence in part (ii) implies algebraic independence over $K_{\partial,0}$ itself. Hence the combined boundary-and-moment field has transcendence degree $d + \mu$ over $k(z)$. Beukers' form of Siegel–Shidlovskii [2, Theorem 1.1, p. 370], applied first to the boundary vector and then to the combined vector, gives the numerical degrees d and $d + \mu$ at every $\xi \in k^\times$. Their difference is μ . The one-variable normal forms have only powers of z as denominators, so these reduced boundary values generate the stated face-value field at all such ξ .

(iv) Specialize (i) at $z = \xi$: the boundary term lies in $K_{\partial,\xi}$, so algebraicity forces all $a_{r,m}(\xi) = 0$ by (iii), which gives the exactness; conversely exactness gives membership by Stokes. $\square$

13 The rank-six example

Let $q = x^4 + y^3 + xy - y$, so $(a, b) = (4, 3)$, $\mu = 6$, $B = \{1, x, x^2, y, xy, x^2y\}$, and D the triangle of Example 1.10. The following are exact computations (script supplied).

- *Connection.* $A = A_0 + A_1/z$ with $\det(t - A_0)$ proportional to

$$\chi(t) = 1289945088 t^6 - 573308928 t^4 + 83980800 t^3 + 83854656 t^2 + 5477429 t - 4197429.$$

The same polynomial is obtained independently as the resultant eliminating the critical points.

- *(M) and the stronger (Q), hence (R).* χ is irreducible over $\mathbb{Q}$; modulo 7 it factors with degrees (1, 5) and modulo 31 with degrees (1, 2, 3). A transitive group of degree 6 containing a 5-cycle is primitive, and with a transposition (the cube of the second Frobenius) it is S_6 . Given a relation $\sum a_i c_i = 0$ with rational a_i , applying a transposition and subtracting yields $(a_i - a_j)(c_i - c_j) = 0$. Thus all a_i are equal; a relation among the differences has coefficient sum zero and is trivial.

- (S). The critical points are $x = 1 - 3y^2$ with $-108y^6 + 108y^4 - 36y^2 + y + 4 = 0$. Write $f(y)$ for this sextic. Sturm counts give exactly two real roots, one in $(-\frac{420}{1000}, -\frac{419}{1000})$ and one in $(\frac{722}{1000}, \frac{723}{1000})$. On the critical locus $\det \text{Hess } q = 72x^2y - 1 = -f'(y)$, and f' has no root in either interval, with $f' > 0$ on the first and $f' < 0$ on the second. So the first root gives a saddle $P_0 \approx (0.471644, -0.419665)$, $c = q(P_0) \approx 0.197305$, and the second does not. The polynomials $x - \frac{y+1}{2}$ and $1 - \frac{y+1}{2} - x$ with $x = 1 - 3y^2$ have no root on the first interval and are positive there, so P_0 is interior to D .
- (F) and the rest of (S) hold by Corollary 1.9 with $k_0 = \mathbb{Q}$, since $[\mathbb{Q}(c) : \mathbb{Q}] = 6 \geq 4$.

For reproducibility, writing $\bar{\chi}_p$ for the monic reduction modulo p ,

$$\begin{aligned}\bar{\chi}_5 &= t^6 - t^4 + 2t^2 - 2t + 2, \\ \bar{\chi}_7 &= (t + 3)(t^5 - 3t^4 + 3t^2 + 3t + 2), \\ \bar{\chi}_{31} &= (t - 14)(t^2 + 8t + 10)(t^3 + 6t^2 - 14t + 1).\end{aligned}$$

The displayed factors modulo 7, 31 are irreducible and squarefree. Irreducibility modulo 5 is certified by

$$\gcd(\bar{\chi}_5, t^{5^2} - t) = \gcd(\bar{\chi}_5, t^{5^3} - t) = 1, \quad t^{5^6} \equiv t \pmod{\bar{\chi}_5}.$$

All these are exact polynomial computations in the accompanying certificate. The supplementary exact checks give $\text{tr}(A_0^k(A_1 + I)) = 0$ for $0 \leq k \leq 5$. Since A_0 has simple spectrum, the Vandermonde matrix in its eigenvalues shows that all six diagonal entries of A_1 in its eigenbasis are -1 . Thus the six powers in Lemma 7.1 are exactly -1 .

The exact checks are run by `make test`. No numerical quadrature or asymptotic fitting is used to certify the hypotheses.

14 The elliptic triangle and the continuation alternative

Fix the counterclockwise triangle

$$D = \text{conv}\{(-1/2, -1), (1/2, 0), (-1/2, 1)\} = \{-1/2 \leq x \leq 1/2, |y| \leq 1/2 - x\},$$

and

$$q = y^2 - x^3 + 3x, \quad M_j = \int_D x^j e^{zq} dx dy \quad (j = 0, 1).$$

A polynomial parametrization is $\Phi(u, v) = (u - 1/2, (1 - u)(2v - 1))$, with Jacobian $2(1 - u)$.

For $\beta = P dy - Q dx$, one has

$$d_z \beta = (L_x P + L_y Q) dx \wedge dy, \quad L_x = \partial_x + 3z(1 - x^2), \quad L_y = \partial_y + 2zy.$$

The operators act in separate variables. Their one-variable quotients have bases $1, x$ and 1 , respectively; both operators are injective on polynomials. Tensoring the two one-variable complexes proves

$$H_q^2 = F_0[dx \wedge dy] \oplus F_0[x dx \wedge dy]. \quad (14.1)$$

In particular the rank is exactly two, not merely at most two.

The explicit reduction recurrences are

$$[x^i y^j] = -\frac{j-1}{2z} [x^i y^{j-2}] \quad (j \geq 1), \quad (14.2)$$

$$[x^{m+2}] = [x^m] + \frac{m}{3z} [x^{m-1}] \quad (m \geq 0), \quad (14.3)$$

with the zero-coefficient terms omitted. Each use records a polynomial primitive. Hence every $r \in k[x, y]$ has a unique remainder $a_r(z) + b_r(z)x$, with $a_r, b_r \in k[z, z^{-1}]$, and an identity

$$r = a_r + b_r x + L_x P_r + L_y Q_r, \quad P_r, Q_r \in k[z, z^{-1}, x, y]. \quad (14.4)$$

Let

$$C = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -5/6 & 0 \\ 0 & -7/6 \end{pmatrix}, \quad A = C + D_0/z.$$

The following are identities of polynomial forms, with Laurent coefficients in z :

$$q \, dx \wedge dy = -\frac{5}{6z} \, dx \wedge dy + 2x \, dx \wedge dy + d_z \beta_0, \quad (14.5)$$

$$q \, dx \wedge dy = 2 \, dx \wedge dy - \frac{7}{6z} x \, dx \wedge dy + d_z \beta_1, \quad (14.6)$$

where

$$\beta_0 = \frac{x}{3z} \, dy - \frac{y}{2z} \, dx, \quad \beta_1 = \frac{x^2 - 2}{3z} \, dy - \frac{xy}{2z} \, dx.$$

Define actual face integrals

$$B_{x,j} = \int_{\partial D} x^j e^{zq} \, dy, \quad B_{y,j} = - \int_{\partial D} x^j y e^{zq} \, dx.$$

Then

$$\begin{pmatrix} M_0 \\ M_1 \end{pmatrix}' = \begin{pmatrix} -5/(6z) & 2 \\ 2 & -7/(6z) \end{pmatrix} \begin{pmatrix} M_0 \\ M_1 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} B_{y,0}/2 + B_{x,1}/3 \\ B_{y,1}/2 + B_{x,2}/3 - 2B_{x,0}/3 \end{pmatrix}. \quad (14.7)$$

This is obtained by integrating (14.6), not by choosing an isomorphic abstract connection and then normalizing a solution.

14.1 A finite actual boundary field

Put

$$p(x) = -x^3 + x^2 + 2x + \frac{1}{4}, \quad P_j(z) = \int_{-1/2}^{1/2} x^j e^{zp(x)} \, dx, \quad Q_0(z) = \int_{-1}^1 e^{z(y^2 - 11/8)} \, dy,$$

and $E_a = e^{-3z/8}$, $E_b = e^{11z/8}$. Parametrizing the three oriented edges gives

$$B_{x,j} = 2P_j - (-1/2)^j Q_0, \quad B_{y,j} = P_j - 2P_{j+1}. \quad (14.8)$$

The one-variable reductions imply

$$K_{\partial,0} = k(z)(P_0, P_1, Q_0, E_a, E_b). \quad (14.9)$$

Indeed the two slanted faces have the same phase p ; their arbitrary polynomial amplitudes reduce to P_0, P_1 and endpoints. The vertical quadratic amplitudes reduce to Q_0 and endpoints.

For completeness, a closed boundary system is

$$P'_0 = \left(\frac{17}{36} - \frac{1}{3z} \right) P_0 + \frac{14}{9} P_1 + \frac{E_b + 5E_a}{18z}, \quad (14.10)$$

$$P'_1 = \left(\frac{28}{27} + \frac{1}{9z} \right) P_0 + \left(\frac{163}{108} - \frac{2}{3z} \right) P_1 + \frac{-53E_b + 41E_a}{108z}, \quad (14.11)$$

$$Q'_0 = \left(-\frac{11}{8} - \frac{1}{2z} \right) Q_0 + \frac{E_a}{z}, \quad E'_a = -\frac{3}{8} E_a, \quad E'_b = \frac{11}{8} E_b. \quad (14.12)$$

For example, $P_2 = (2P_1 + 2P_0)/3 - (E_b - E_a)/(3z)$, so (14.7) closes with this boundary vector. The only finite pole of the combined rational system is zero.

14.2 The actual normalization at zero

The face forcing in (14.7) is $b(z)/z$ with b entire. Writing $M = \sum v_m z^m$, $b = \sum b_m z^m$, its holomorphic solution obeys

$$(mI - D_0)v_m = Cv_{m-1} + b_m, \quad v_{-1} = 0.$$

Every $mI - D_0$, $m \geq 0$, is invertible. Thus there is at most one holomorphic solution at zero with these actual forcing functions. The integral supplies it, with

$$M_0(0) = 1, \quad M_1(0) = -\frac{1}{6}.$$

This supplies a second, independent check that no unspecified homogeneous solution has been added to the actual column.

14.3 The actual continuation

On the regular phase fibre $y^2 = x^3 - 3x + t$, the convention $dq \wedge \alpha = dx \wedge dy$ gives $\alpha = -dx/(2y)$.

Lemma 14.1 (Continuation for the elliptic triangle). *The density germ of M_0 just to the right of $t = 0$ satisfies (S^*) .*

Proof. We verify a nonzero elliptic monodromy variation of the actual density germ. Put $f(x) = -x^3 + 3x$, let $r_2(s) \in (-1, 1)$ be its middle inverse branch for $-2 < s < 2$, and let $c(t)$ be the inverse branch of p beginning at $c(0) \in (-1/5, 0)$. Set $a(t) = 1/2 - c(t)$. Coarea on the triangle gives, near zero,

$$\rho(t) = \int_{-a(t)}^{a(t)} \frac{dy}{3(1 - r_2(t - y^2)^2)}. \quad (14.13)$$

Here $p'(x) = -3x^2 + 2x + 2$ is positive on $[-1/5, 1]$. Moreover $p(-1/5) < 0 < p(0)$ and $p(4/5) < 2 < p(1)$, so $c(t)$ is a single real-analytic branch for $0 \leq t \leq 2$, with $c(0) \in (-1/5, 0)$ and $c(2) \in (4/5, 1)$. For $|y| \leq |a(t)|$ one has

$$f(c(t)) = t - a(t)^2 \leq t - y^2 \leq t < 2, \quad f(c(t)) > -2.$$

Thus the middle inverse branch is regular on this moving segment for $0 < t < 2$. Writing the signed integral on the fixed interval as

$$a(t) \int_{-1}^1 \frac{ds}{3(1 - r_2(t - a(t)^2 s^2)^2)}$$

proves real analyticity and hence local holomorphic continuation along that interval. At $t = 11/8$ one has $c(t) = 1/2$, so the endpoints meet, but the displayed expression is analytic and zero there. This specifies the continuation through that otherwise excluded boundary value. Near $t = 2$, $a(t) < 0$ and $a(2) \neq 0$, and both endpoints are unramified.

Write $\varepsilon = 2 - t$ and $r_2(t - y^2) = 1 - v$. Since $f(1 - v) = 2 - 3v^2 + v^3$, near the node the integrand is

$$\frac{1}{2\sqrt{3}\sqrt{\varepsilon + y^2}} + O(1).$$

Choose a fixed $b > 0$ smaller than $|a(2)|$. The outer parts $|y| \geq b$ extend holomorphically to $t = 2$; on $|y| \leq b$ the remainder in the displayed expansion is uniformly bounded for $\varepsilon > 0$. Since the integration orientation is reversed near 2, it follows that

$$\rho(t) = \frac{1}{2\sqrt{3}} \log(2 - t) + O(1) \quad (t \rightarrow 2^-). \quad (14.14)$$

The local geometry makes the corresponding variation precise. Put $v = 1 - x$ and $u = v\sqrt{3 - v}$. Then $q - 2 = y^2 - u^2$ near the node, so a nearby local fiber is an annulus. The path outside a small nodal neighborhood and its two endpoints return under a sufficiently small loop. Its variation lies in the one-dimensional cycle space of that annulus. The coefficient in (14.14) proves it is nonzero. The local core cycle δ_+ is fixed by this nodal monodromy (equivalently by its Dehn twist). Consequently

$$(T_+ - I)\rho = c_* h_+(t), \quad c_* \neq 0, \quad h_+(t) = \int_{\delta_+} \frac{dx}{2y}, \quad T_+ h_+ = h_+.$$

This is also the local annulus proof of the Picard–Lefschetz formula in this case; the nonzero coefficient has just been checked on the actual path by (14.13).

The variation is a nonzero constant multiple of the required Gelfand–Leray vanishing period; the common sign is immaterial. $\square$

Remark 14.2. This is continuation of a regular density germ, not a claim that the original compactly supported density is nonzero outside $q(D) = [-11/8, 11/8]$.

14.4 Verification of the comparison hypotheses

The phase has $(a, b) = (3, 2)$ and its two critical values are $-2, 2$. Thus (M) holds and (Q) is automatic. For a slanted edge put $\beta = 7\sqrt{7}/27$. The substitution $x = 1/3 + (\sqrt{7}/3)s$ gives

$$p(x) = \frac{107}{108} + \beta(-s^3 + 3s).$$

Its critical values are $107/108 \pm 2\beta$, so their nonzero differences are $\pm 4\beta$, whereas the interior differences are ± 4 . Since $\beta^2 = 343/729 \neq 1$, these do not agree. The vertical quadratic face has only one critical value, so it also satisfies (F). Lemma 14.1 supplies (S^*) , although the saddle $(1, 0)$ lies outside the triangle and (S) fails.

Corollary 14.3 (Elliptic functional and numerical comparison). *For this triangle and phase, M_0, M_1 are algebraically independent over $\mathbb{C}(z)(P_0, P_1, Q_0, E_a, E_b)$. At every nonzero algebraic ξ , their values are algebraically independent over the corresponding fixed-face value field. Every polynomial amplitude has the unique remainder (14.4), and its value is algebraic over the face-value field exactly when its specialized polynomial twisted class is zero.*

Proof. Apply Theorem 1.7, using the verifications above. The formulas (14.6)–(14.9) retain the actual normalization and the finite generating boundary vector. Thus this is also a specialization of the general proof, not just an abstract identification of two homogeneous equations. $\square$

15 A noncoprime rank-three comparison

The preceding geometric irreducibility proof uses coprime weights, but coprimality is not necessary for every individual exponential module. Consider

$$q = x^4 + y^2 + x, \quad T = \text{conv}\{(-1, -1), (0, -\tfrac{1}{2}), (-\tfrac{1}{2}, 1)\}.$$

$$U_j(z) = \int_T x^j e^{zq} dx dy \quad (0 \leq j \leq 2).$$

Use the positive orientation and only the three specified edges.

Theorem 15.1. *The three functions U_0, U_1, U_2 are algebraically independent over the fixed-face field K_∂ . For every $\xi \in k^\times$ their values are algebraically independent over $K_{\partial, \xi}$. Every polynomial amplitude has a unique Laurent remainder in $1, x, x^2$, and its integral at ξ is algebraic over $K_{\partial, \xi}$ if and only if its two-form is polynomially $(d + \xi dq \wedge)$ -exact.*

15.1 Literal system and formal exponents

The proof of Lemma 5.1 does not use coprimality. Applied to $(a, b) = (4, 2)$, it gives the free basis $1, x, x^2$ and a system

$$U' = AU + b, \quad A = C + \frac{D_0}{z}, \quad C = \begin{pmatrix} 0 & 3/4 & 0 \\ 0 & 0 & 3/4 \\ -3/16 & 0 & 0 \end{pmatrix}, \quad D_0 = \text{diag}(-3/4, -1, -5/4). \quad (15.1)$$

In fact for $j = 0, 1, 2$ the polynomial certificates are

$$qx^j = \sum_{l=0}^2 A_{jl}x^l + L_x P_j + L_y Q_j, \quad P_j = \frac{x^{j+1}}{4z} + \frac{3\delta_{j,2}}{16z}, \quad Q_j = \frac{x^j y}{2z}.$$

Thus $b_j = \int_{\partial T} e^{zq} (P_j dy - Q_j dx)$ is the actual boundary forcing. No choice of a homogeneous solution normalizes this column.

The critical points satisfy $4x^3 + 1 = 0$, $y = 0$, and their critical values $c_i = 3x_i/4$ are the simple roots of

$$c^3 + \frac{27}{256} = 0.$$

They have six distinct ordered differences. A nontrivial equality among three distinct values would make one the midpoint of the other two; since their sum is zero this would force a zero critical value.

For the spectral projector onto the c_i -eigenline of C ,

$$\Pi_i = \frac{1}{3} \left(I + \frac{C}{c_i} + \frac{C^2}{c_i^2} \right), \quad \text{tr}(\Pi_i D_0) = -1,$$

because $\text{tr}(CD_0) = \text{tr}(C^2 D_0) = 0$ and $\text{tr}(D_0) = -3$. Lemma 7.1 therefore gives the three formal exponents -1 at infinity.

15.2 Direct irreducibility

Lemma 15.2. *The differential module of (15.1) is irreducible over $\mathbb{C}(z)$, and its homogeneous differential Galois group is SL_3 .*

Proof. A rational invariant line for the solution connection would be generated by a primitive polynomial vector $v(z)$, whose entries have no common zero. Write its scalar connection as $r(z)$, so $v' + rv = Av$. At an ordinary finite point one entry of v is nonzero, hence r has no pole there; at zero it has at most a simple pole. Using an entry of maximal degree shows $r = O(1)$ at infinity. Thus $r = c + \lambda/z$. A corresponding solution is $e^{cz} z^\lambda v(z)$.

At zero $v(0) \neq 0$, so $\lambda \in \{-3/4, -1, -5/4\}$. Let $d = \deg v$. Comparison of the leading and next coefficients at infinity, or the formal diagonalization above, gives $\lambda + d = -1$. With d a nonnegative integer the only possibility is $\lambda = -1$, $d = 0$. But the -1 eigenspace of D_0 is the middle coordinate line, and C does not preserve it. There is no such invariant line.

For the dual system $-A^T$, the residue eigenvalues are $3/4, 1, 5/4$ and the formal exponents are all 1. The same comparison forces $\lambda = 1, d = 0$, but the middle line is not preserved by $-C^T$ either. Thus there is no invariant line in the dual. A proper submodule in rank three has rank one or produces a rank-one quotient; both have been excluded. This proves irreducibility, independently of any assertion about the punctures of an ordinary fibre.

Proposition 7.2 now gives $G \supseteq \mathrm{SL}_3$. Since $\mathrm{tr} A = -3/z$, a determinant solution is $z^{-3} \in F$, so $G \subseteq \mathrm{SL}_3$ as well. $\square$

15.3 The triangle and its boundary

The unique global real minimum is $(-4^{-1/3}, 0)$. Since $1/2 < 4^{-1/3} < 3/4$, it is strictly inside T : the three bounding lines are

$$y = x/2 - 1/2, \quad y = -3x - 1/2, \quad y = 4x + 3.$$

The minimum is unique and the boundary compact, so (E) holds strictly. Lemma 3.2 supplies actual detection.

For a quartic $p = x^4 + Ax^2 + Bx + C_0$, let u_1, u_2, u_3 be its critical values with multiplicity. Multiplication by p in $k[x]/(p')$, or Newton identities, gives

$$\sum_{i \neq j} (u_i - u_j)^2 = 6 \mathrm{tr}(m_p^2) - 2 \mathrm{tr}(m_p)^2 = \frac{A(A^3 - 27B^2)}{4}. \quad (15.2)$$

On an edge $y = mx + b_0$, take $A = m^2, B = 1 + 2mb_0$. The three values of (15.2), in the order of the lines above, are

$$-\frac{431}{1024}, \quad \frac{2673}{4}, \quad -51116.$$

For the interior cubic critical-value polynomial the same ordered-difference square sum is zero. An edge has at most six ordered critical differences. Containing the six distinct interior differences would force equality of the two sets, without repetitions, and equality of their square sums. This is impossible, so (F) holds on each edge, whether or not its own critical values repeat.

Proof of Theorem 15.1. Apply Theorem 4.3 using Lemma 15.2, Proposition 9.4, and the extremum detector. This gives functional independence. The actual combined system has coefficients in $k[z, z^{-1}]$ and E -function coordinates. Applying Beukers' theorem separately to the boundary and combined vectors, exactly as in the proof of Theorem 1.7(iii), gives independence at every $\xi \neq 0$. Specialized Laurent reduction and Stokes give the exactness equivalence. $\square$

This is an explicit noncoprime extension, not a theorem for all noncoprime weighted phases. The calculations are reproduced by the comparison and factorial-moment verification script.

16 Scope, dependencies, and verification

The normal-form theorem is algebraic and the polynomial boundary-system construction allows arbitrary polynomial phases. Linear comparison requires irreducibility and actual detection; for the coprime weighted class, (M) and one of (S), (S*), (E) suffice. Full independence needs the additional root and boundary hypotheses of Theorem 1.7. Finite root separation replaces the older arithmetic hypothesis (Q); Proposition 8.1 proves its nonempty Zariski-open locus in the full weighted family, without claiming generic real-chain detection. The noncoprime rank-three application uses a

separate direct irreducibility proof. Its scope is not a universal noncoprime theorem. All boundary fields are attached to the finitely many fixed faces and endpoints of the chain. Replacing them by the field of all compact polynomial-line periods is not an asserted extension. Only one phase and one actual column are considered. No several-chain or multiblock formal completeness is inferred. Numerical independence is a statement after specializing the parameter; it is not injectivity of evaluation on the unspecialized algebra containing $z - \xi$.

External inputs. The geometric proof uses Broughton [4, Definition 3.1, Theorem 1.2, pp. 229–231] and the Picard–Lefschetz, distinguished-basis, connectedness, and nondegeneracy statements in [1, §1.3; §2.1; Theorem 3.3; pp. 100, 411]. The needed perturbation and compatible-transport arguments are included in Proposition 6.1. Polynomial relative exactness uses [3, Definitions 1.3–1.4, Theorem 1.5]. Ordinary Picard–Vessiot theory and its tensor equivalence use [11, Theorem 1.5.2], and the complete reducibility in the actual criterion uses [7, Proposition 15.15, Corollary 22.43]. The general finiteness assertion uses the explicitly identified Proposition 9.1 of Sabbah [10]. Linear specialization uses [2, Theorem 3.2]; full independence uses [2, Theorem 1.1] separately for the boundary and combined vectors. Neither Wasow’s formal simplification nor a general formal-normal-form theorem is needed: the relevant simple-spectrum and lattice statements are proved in the text.

Computational verification. The repository contains the retained exact elliptic-amplitude certificate, the rank-six polynomial/resultant/Frobenius/Sturm certificate, and exact checks of the formal recursion and six formal powers. Additional exact tests verify the strict (R)-versus-(Q) example, the rank-three Stokes identities and three edge spectra, and the characteristic-five application formulas. The rank-six geometric inequalities are certified on whole rational intervals. These tests verify finite calculations, not the general monodromy or Galois proofs. Compilation, regression tests, and AI-assisted audits are not independent specialist refereeing or formal verification.

AI assistance and responsibility. OpenAI ChatGPT and Anthropic Claude were used for proof exploration, drafting, symbolic checks, and adversarial review. They are not authors. The authors are responsible for the statements, proofs, citations, and remaining errors.

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